Group - III crossection derivation (corrected)

Corrections (blue; checked against Calculation_main.tex, see Groups_II_III_check.pdf):

- Rows III, V, VI: the overall sign. In Eq. (CS2) of Calculation_main.tex these rows enter as $-4\,\mathfrak{C}'^{\dagger}\mathfrak{B}$, $-2\,\mathfrak{C}'^{\dagger}\mathbb{X}'_1$, $-2\,\mathfrak{C}'^{\dagger}\mathbb{X}'_2$, because the splitting amplitude after the target is $\mathfrak{C} = -\int_y[\bar{\mathbb{C}}U^{\dagger}U\mathbb{A} - U^{\dagger}\bar{\mathbb{C}}\bar{\mathbb{A}}]$ ($\bar{\mathbb{C}}_2^{\dagger} = -\bar{\mathbb{C}}_1$, $\mathbb{A}_1^{\dagger} = -\mathbb{A}_2$). Every line of these rows changes sign, including the $\log(\vee/\Lambda)$ terms.
- Plus prescription: the lower limit of the p^+ integral stays Λ (exact); with 0 the result is only approximate.
- Missing or extra integration variables in the measures.
- Not changed here: index placements of the adjoint Wilson lines. They are correct with your convention $U^{bc} = U^{cb}$; for a general adjoint U ($U^{\dagger} = U^T \neq U$) see Sec. 5 of the check note.

1 Group III (with c.c.)

1.1 Row I [Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{2}{(2\pi)^3}\int_{\Lambda}^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\left[\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')\bar{\mathbb{A}}_j^{\dagger\,(1)\,c'}(p^+,-z)U^{c'b}(z)\right.$$
$$\left. - \mathbb{A}_i^{\dagger\,(1)\,c'}(k^+,-w')U^{bd'}(z)\bar{\mathbb{A}}_j^{\dagger\,(1)\,d'}(p^+,-z)U^{c'a}(w')\right]\times\left[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\right]$$

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4}{4\pi^4}\int_{\Lambda}^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}$$
$$\left[\frac{\delta^{ij}\,\left[(x-z)^2-(x-w)^2\right]}{2\,(p^++k^+)\,(z-w)^2} + \frac{1}{p^+}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2}\right] + \frac{1}{k^+}\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2}\right.\right.$$
$$\left.\left. - \frac{(x-z)^j\,(x-w)^i}{2\,(x-w)^2}\right]\right]\times\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{1.1}$$

Integrating w.r.t. p^+ and simplifying we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\left\{\right.$$
$$-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{k^+\,(x-w)^2+\Lambda\,(x-z)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2} + \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left. - \frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2} - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]$$
$$+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^++\Lambda}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}$$
$$-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\left.\right\}\times$$
$$\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{1.2}$$

1.2 Row II [Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{2}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\left[\bar{\mathbb{A}}_j^{\dagger\,(1)\,c'}(p^+,-z)\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')U^{bc'}(z) - \mathbb{A}_j^{\dagger\,(1)\,b}(p^+,-z)\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')\right]\times$$
$$\left[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\right]$$

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\frac{1}{\left[p^+\,(x-z)^2+k^+\,(x-w)^2\right]}$$
$$\left[\frac{\delta^{ij}\,\left[(x-z)^2-(x-w)^2\right]}{2\,(p^++k^+)\,(z-w)^2}+\frac{1}{p^+}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2}\right]+\frac{1}{k^+}\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2}\right.\right.$$
$$\left.\left.-\frac{(x-z)^j\,(x-w)^i}{2\,(x-w)^2}\right]\right]\times\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{1.3}$$

Integrating w.r.t. p^+ and simplifying we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\left\{\right.$$
$$-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{k^+\,(x-w)^2+\Lambda\,(x-z)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\right]$$
$$+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^++\Lambda}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}$$
$$-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times$$
$$\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{1.4}$$

1.3 Row III [Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{4}{(2\pi)^3}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y'}\left[\mathbb{A}_{k'}^{\dagger\,(1)\,e'}(p^++k^+,-y')U^{\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)\right.$$
$$\left.-\bar{\mathbb{A}}_{k'}^{\dagger\,(1)\,c'}(p^++k^+,-y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)U^{d'b}(z)\right]\times\left[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z)\right.$$
$$\left.-U^{\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\right]$$

We write it in two parts:

1.3.1 First part: [Done ✓]

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = +\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^4}\,\frac{1}{k^{+\,2}}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',x',y',z,y,x} e^{-ik\cdot(w'-w)}\frac{(w'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(w'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\delta^{(2)}\left[w'-\frac{(p^++k^+)\,y'-p^+\,z}{k^+}\right]\times\left[\delta_{k'm}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm}\,\delta_{ik'}-\frac{(p^++k^+)}{k^+}\,\delta_{im}\,\delta_{jk'}\right]$$
$$\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \tag{1.5}$$

Integrating w.r.t. w'

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = + \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^4}\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \frac{1}{p^+}\,\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \tag{1.6}$$

Using + prescription, we get Here $\int_\Lambda^{\vee-k^+} dp^+\left[\frac{1}{p^+}\right]_+ g(p^+) \equiv \int_\Lambda^{\vee-k^+} dp^+\,\frac{g(p^+)-g(0)}{p^+}$, so that $\int_\Lambda^{\vee-k^+}\frac{dp^+}{p^+}g(p^+) = \int_\Lambda^{\vee-k^+} dp^+\left[\frac{1}{p^+}\right]_+ g(p^+) + g(0)\log\frac{\vee-k^+}{\Lambda}$ exactly. The lower limit stays Λ for all terms: extending it to 0 drops $\int_0^\Lambda dp^+(\dots)$, which is not small unless $\Lambda \ll k^+$ and $\Lambda|\mathbf{k}||y'-z| \ll k^+$.

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = + \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^4}\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left[\frac{1}{p^+}\right]_+\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]$$
$$-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{16\pi^5}\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right] + \log\left[1-\frac{k^+}{\vee}\right]\right]\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \tag{1.7}$$

1.3.2 Second part: [Done ✓]

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = + \frac{1}{(2\pi)^3}\frac{g^4}{4\pi^4}\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',y',z,y,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-y')^{k'}}{(w'-z)^2\,(x'-y')^2}\,\frac{p^+}{(p^+ + k^+)\,[p^+\,(x-z)^2 + k^+\,(x-w)^2]}$$
$$\delta^{(2)}\left[\frac{p^+}{k^+}\,(y'-z) + y' - w'\right]\times\left[\frac{\delta_{k'm}\,d_\perp}{2\,(z-w)^2}\,\left[(x-z)^2-(x-w)^2\right] + \frac{(p^+ + k^+)\,\delta_{k'm}}{k^+}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}\right.\right.$$
$$-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2} - \left.\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right] + \frac{(p^+ + k^+)\,\delta_{k'm}}{p^+}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2} + \frac{(x-z)\cdot(x-w)}{2\,(x-z)^2}\right.$$
$$-\left.\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right] - \frac{(p^+ + k^+)^2}{p^{+\,2}}\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2} + \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right] - \frac{(p^+ + k^+)^2}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}\right.$$
$$-\left.\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right] - \frac{(p^+ + k^+)^2}{p^+\,k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2} - \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2} + \frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2}\right.$$
$$\left.\left.+\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\right]\times\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{1.8}$$

Integrating w.r.t w'

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = + \frac{1}{(2\pi)^3}\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2 + k^+\,(x-w)^2]}$$
$$\left[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^+ + k^+)^2}\,\left[(x-z)^2-(x-w)^2\right] + \frac{p^+\,\delta_{k'm}}{k^+\,(p^+ + k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2} - \frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.\right.$$
$$-\left.\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right] + \frac{\delta_{k'm}}{(p^+ + k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2} + \frac{(x-z)\cdot(x-w)}{2\,(x-z)^2} - \frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]$$
$$-\frac{1}{p^+}\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2} + \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right] - \frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]$$
$$-\frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2} - \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2} + \frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2} + \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\Bigg]\times$$
$$\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{1.9}$$

Integrating w.r.t. p^+, using the plus prescription, we get Here $\int_\Lambda^{\vee-k^+} dp^+\left[\frac{1}{p^+}\right]_+ g(p^+) \equiv \int_\Lambda^{\vee-k^+} dp^+\,\frac{g(p^+)-g(0)}{p^+}$, so that $\int_\Lambda^{\vee-k^+}\frac{dp^+}{p^+}g(p^+)$ $\int_\Lambda^{\vee-k^+} dp^+\left[\frac{1}{p^+}\right]_+ g(p^+) + g(0)\log\frac{\vee-k^+}{\Lambda}$ exactly. The lower limit stays Λ for all terms: extending it to 0 drops $\int_0^\Lambda dp^+(\dots)$, which

is not small unless $\Lambda \ll k^+$ and $\Lambda|\mathbf{k}||y'-z| \ll k^+$.

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\\
&\left[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^++k^+)^2}\,\left[(x-z)^2-(x-w)^2\right]+\frac{p^+\,\delta_{k'm}}{k^+\,(p^++k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.\right.\\
&\left.-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]+\frac{\delta_{k'm}}{(p^++k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2}+\frac{(x-z)\cdot(x-w)}{2\,(x-z)^2}-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]\\
&-\left[\frac{1}{p^+}\right]_+\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2}+\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right]-\frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\\
&\left.-\frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2}-\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2}+\frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2}+\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\right]\times\\
&\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{w,x',y',z,y,x}e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.\\
&\left.+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{1.10}
$$

1.4 Row IV

[Done ✓]

Only 4 ρ's

1.5 Row V

[Done ✓]

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = &-\frac{2}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'}e^{-ik(w'-w)}\int_{y'}\left[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^++k^+,-y')U^{\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)\right.\\
&\left.-\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^++k^+,-y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)U^{d'b}(z)\right]\times\left[U^{bc}(z)\bar{\mathbb{A}}^{(1)\,c}_j(p^+,-z)\bar{\mathbb{A}}^{(1)\,a}_i(k^+,-w)\right.\\
&\left.-U^{ac}(w)\bar{\mathbb{A}}^{(1)\,d}_j(p^+,-z)U^{db}(z)\mathbb{A}^{(1)\,c}_i(k^+,-w)\right]
\end{aligned}
$$

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = &+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^{+\,2}}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',y',z,y,x,x'}e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(w'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\delta^{(2)}\left[w'-\frac{(p^++k^+)\,y'-p^+\,z}{k^+}\right]\times\left[\delta_{k'm}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm}\,\delta_{ik'}-\frac{(p^++k^+)}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\\
&\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{bc}(z)\,U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,U^{cb}(z)\,\rho^e(x)\rho^d(y)\right]
\end{aligned}
\tag{1.11}
$$

Integrating w.r.t. w'

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = &+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\frac{1}{p^+}\,\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[U^{bc}(z)\,U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,U^{cb}(z)\,\rho^e(x)\rho^d(y)\right]
\end{aligned}
\tag{1.12}
$$

Using + prescription we obtain the following Here $\int_\Lambda^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+g(p^+)\equiv\int_\Lambda^{\vee-k^+}dp^+\,\frac{g(p^+)-g(0)}{p^+}$, so that $\int_\Lambda^{\vee-k^+}\frac{dp^+}{p^+}g(p^+)=\int_\Lambda^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+g(p^+)+g(0)\log\frac{\vee-k^+}{\Lambda}$ exactly. The lower limit stays Λ for all terms: extending it to 0 drops $\int_0^\Lambda dp^+(\dots)$, which

is not small unless $\Lambda \ll k^+$ and $\Lambda|\mathbf{k}||y'-z| \ll k^+$.

$$\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = &+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left[\frac{1}{p^+}\right]_+\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\big[U^{bc}(z)\,U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,U^{cb}(z)\,\rho^e(x)\rho^d(y)\big]\\
&-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{\vee-k^+}{\Lambda}\right]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{bc}(z)\,U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,U^{cb}(z)\,\rho^e(x)\rho^d(y)\Big]
\end{aligned}\tag{1.13}$$

Simplifying the Wilson lines we get

$$\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = &+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{4\pi^4}\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left[\frac{1}{p^+}\right]_+\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\Big[U^{e'c'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big]\\
&-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\Big[U^{e'c'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big]
\end{aligned}\tag{1.14}$$

1.6 Row VI

[Done ✓]

$$\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = &-\frac{2}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'}e^{-ik(w'-w)}\int_{y'}\Big[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^++k^+,-y')U^{\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)\\
&-\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^++k^+,-y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Big]\times\Big[U^{bc}(z)\bar{\mathbb{A}}^{(1)\,a}_i(k^+,-w)\bar{\mathbb{A}}^{(1)\,c}_j(p^+,-z)\\
&-\bar{\mathbb{A}}^{(1)\,a}_i(k^+,-w)\mathbb{A}^{(1)\,b}_j(p^+,-z)\Big]
\end{aligned}$$

$$\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = &+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^{+\,2}}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',y',z,y,x,x'}e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(w'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\delta^{(2)}\left[w'-\frac{(p^++k^+)\,y'-p^+\,z}{k^+}\right]\times\left[\delta_{k'm}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm}\,\delta_{ik'}-\frac{(p^++k^+)}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\\
&\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x)-U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\Big]
\end{aligned}\tag{1.15}$$

Integrating w.r.t. w'

$$\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = &+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\frac{1}{p^+}\,\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x)-U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]
\end{aligned}\tag{1.16}$$

Using + prescription we obtain the following: Here $\int_\Lambda^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+g(p^+)\equiv\int_\Lambda^{\vee-k^+}dp^+\,\frac{g(p^+)-g(0)}{p^+}$, so that $\int_\Lambda^{\vee-k^+}\frac{dp^+}{p^+}g(p^+)=\int_\Lambda^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+g(p^+)+g(0)\log\frac{\vee-k^+}{\Lambda}$ exactly. The lower limit stays Λ for all terms: extending it to 0 drops $\int_0^\Lambda dp^+(\dots)$, which

is not small unless $\Lambda \ll k^+$ and $\Lambda|\mathbf{k}||y'-z| \ll k^+$.

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = &+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}\,(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left[\frac{1}{p^+}\right]_+\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x)-U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x)-U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]
\end{aligned}
\tag{1.17}
$$